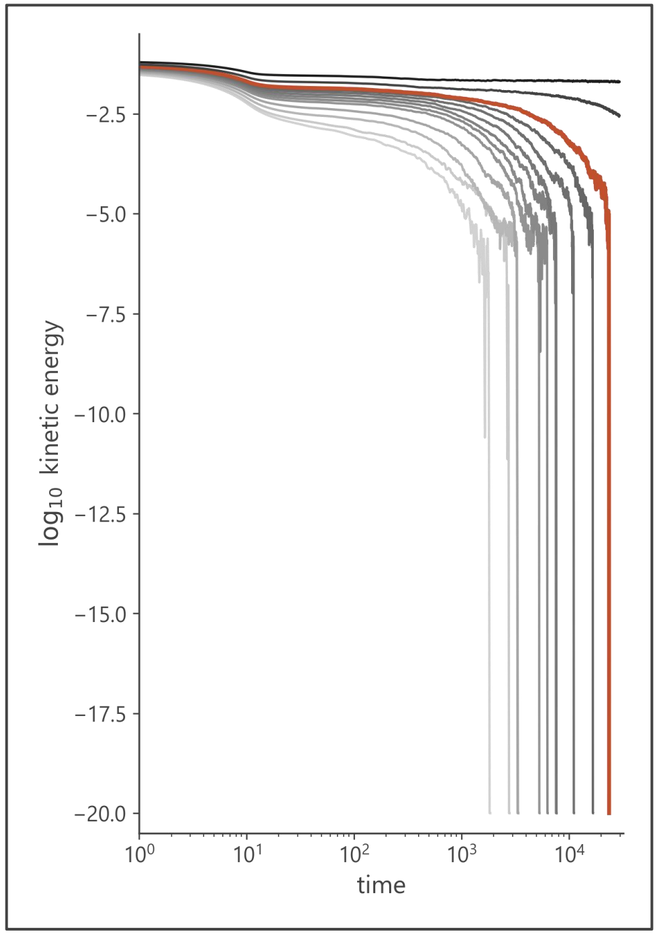
Team B-9

October 2026

Decoding Dense Active Fluids Through Machine Learning

Scaling, Koopman prediction and reduced equations

for kinetic arrest in a dense active glass



Presented by :
Team B-9
Rithvik Arulprakash · Harshith KV · Vipin Sudhakar · Vijay Murugan

Code & data :
github.com/Rithvik188/Compflu

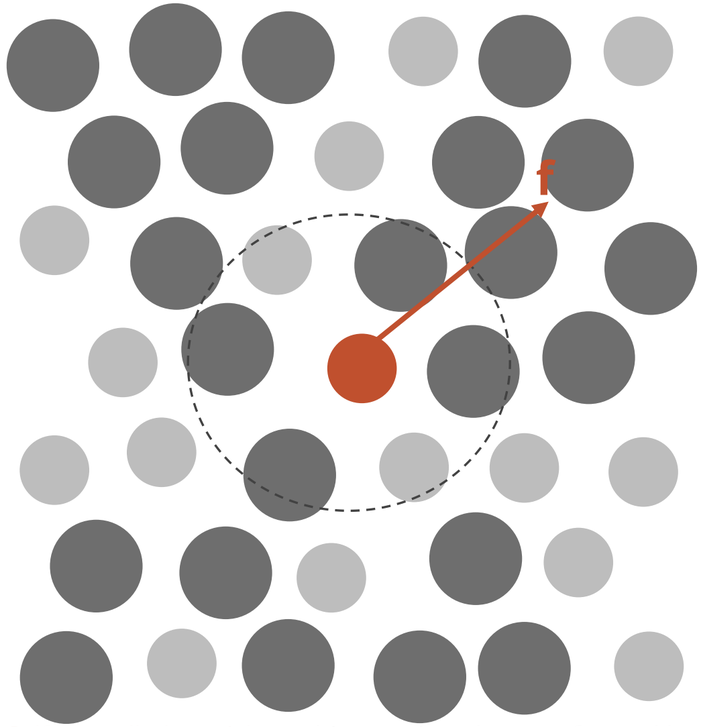
Flow or freeze?

Dense self-propelled particles each push themselves with an **active force f**. Packed tightly, every particle is caged by its neighbours.

Strong pushing keeps the system **flowing**; weak pushing lets it **jam and freeze**. How the motion dies out near this transition has no closed theory.

Our approach

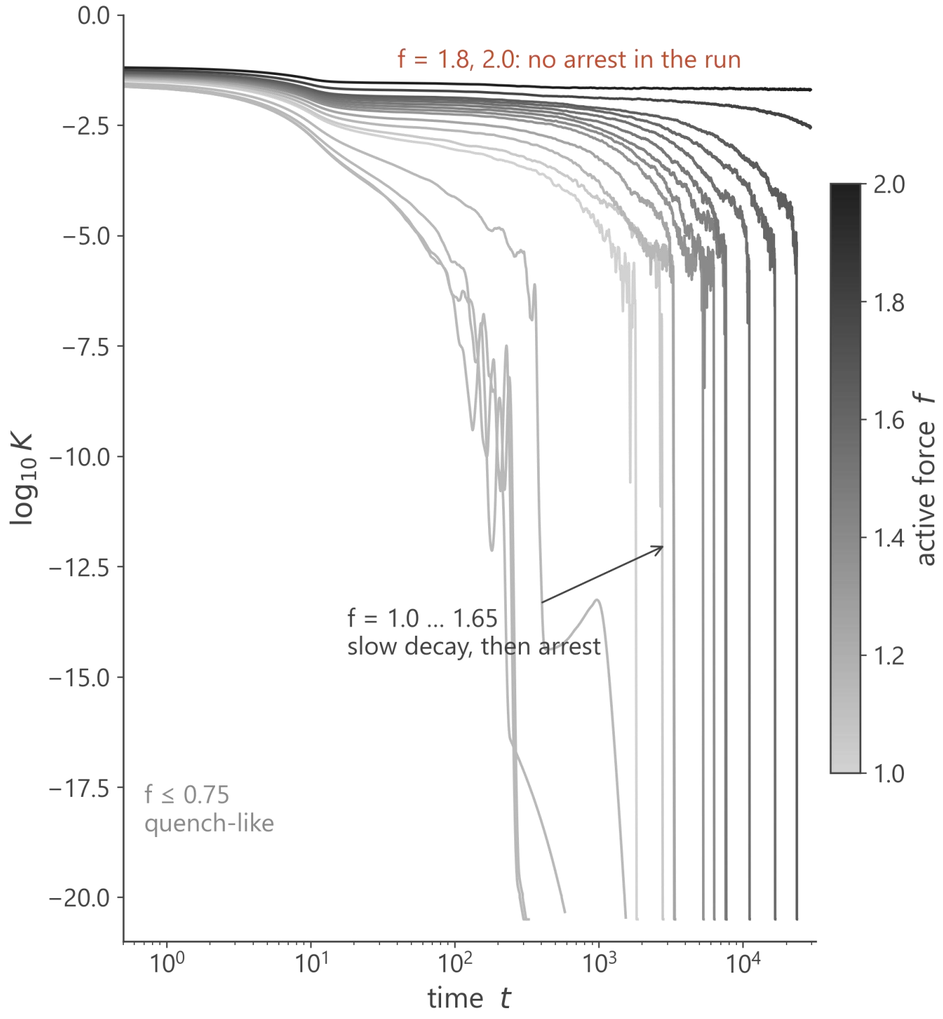
Learn the dynamics of the kinetic energy K(t) directly from simulations: scaling and master curves, Koopman-operator models, HAVOK and sparse equation discovery.



Self-propelled particle pushing against the cage formed by its neighbours (65 : 35 mixture).

System & data

Model	2D Kob–Andersen mixture, 65 : 35, Lennard-Jones
Particles	N = 1000, constant active force f on each
Forces	17 values, f = 0 … 2.0, one simulation each
Signal	mean kinetic energy K(t), Δt = 0.5, up to t ≈ 3×10^4
We study	$\log_{10}$K: K falls by more than 20 decades



Two questions, one pipeline

Q1
Scaling

Do all trajectories follow one master curve, and how does the time scale grow with f?

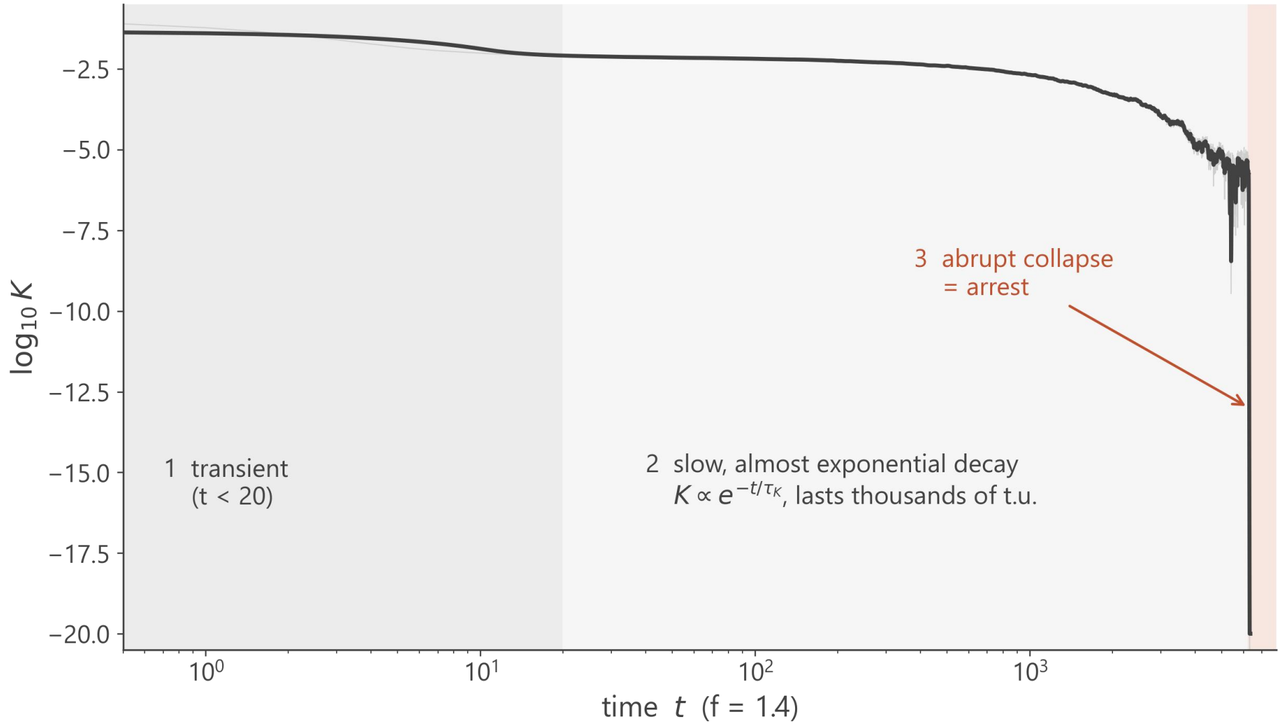
Q2
Prediction

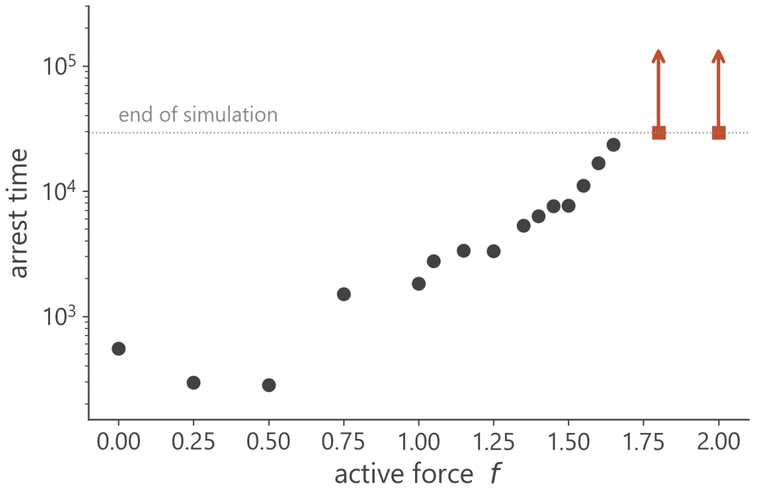
Can Koopman-operator models forecast a K(t) trajectory they have never seen?



Every claim is checked with a test that could fail: baselines, held-out forces, burst-free surrogates and bootstrap intervals.

Anatomy of an arrest



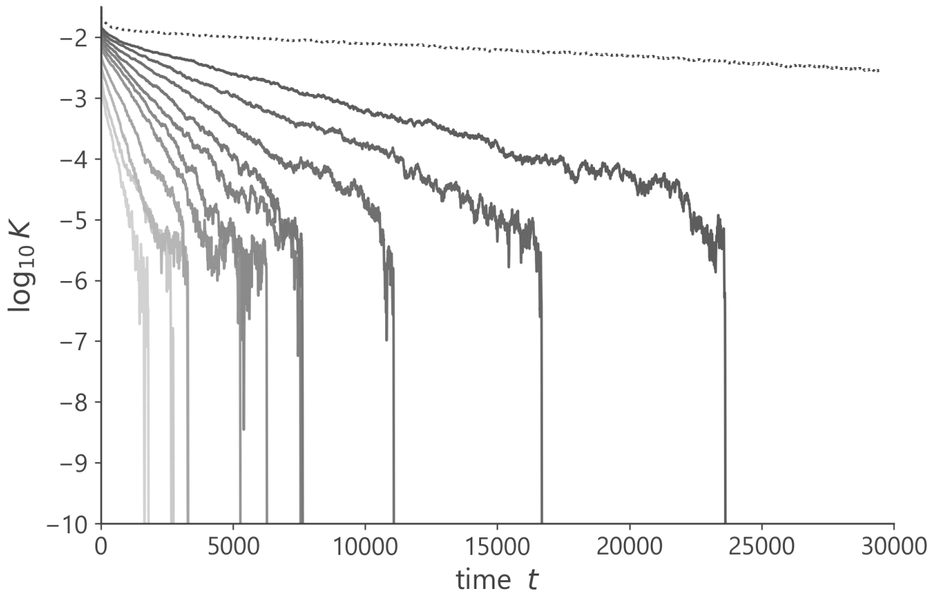


f = 1.65 still arrests (t ≈ 23 650).

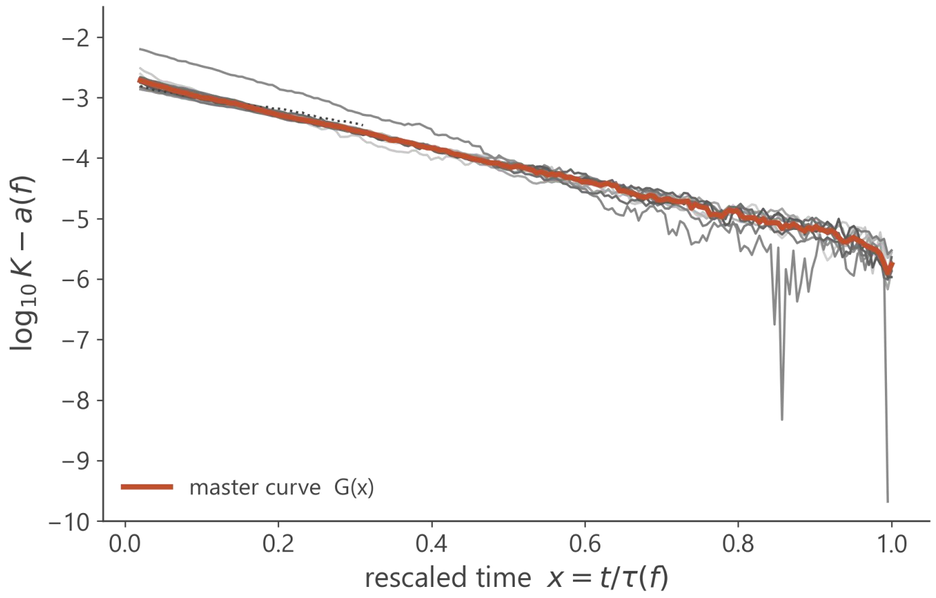
f = 1.8 is still decaying at the end of the run (p < 0.001) → $f_c > 1.8$

Q1 · One master curve

$$\log_{10} K(t, f) = a(f) + G(t / \tau(f))$$



rescale



spread 1.81

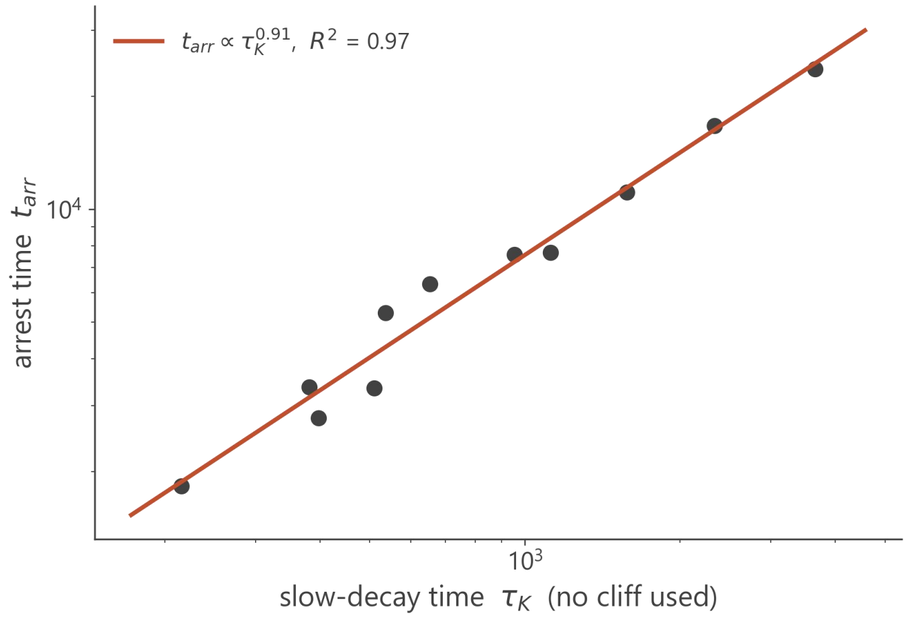
variance of log K across forces, as simulated

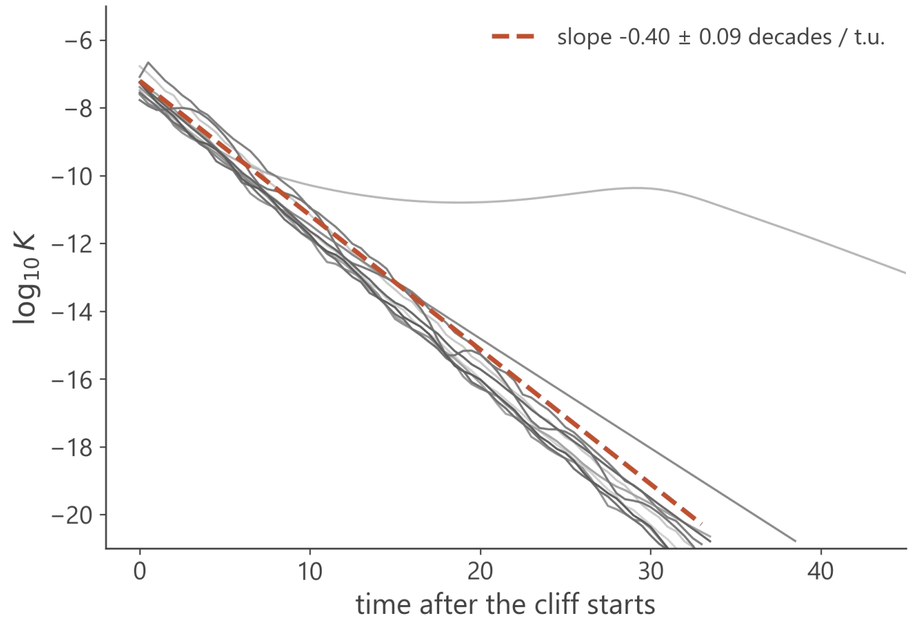
spread 0.030

after $t \to t/\tau(f)$ and a shift $a(f)$ · $R^2 = 0.983$

Is the collapse real?

Two free numbers per curve is a lot of freedom, so we tested what the collapse never used.

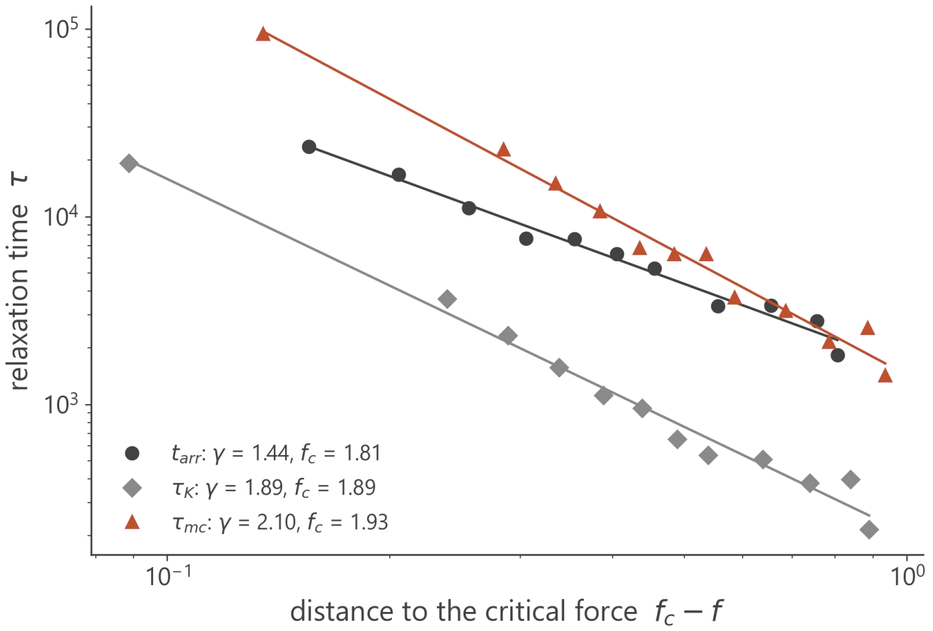


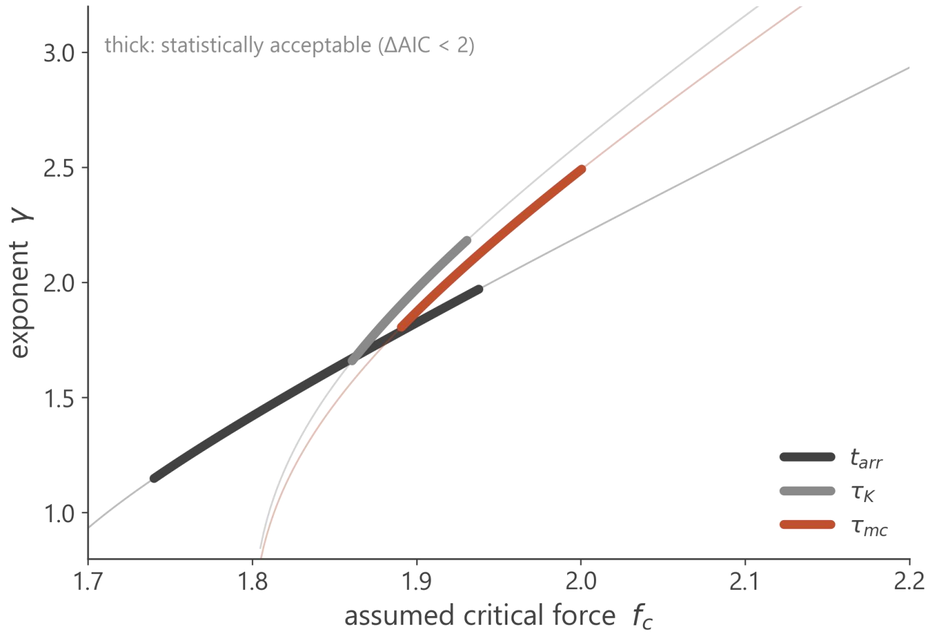


Arrest time follows from the slow decay alone: τ_K varies 17×, the ratio t_{arr}/τ_K only **±15 %**.

After arrest every force collapses at the same rate: **0.40 ± 0.09** decades per time unit.

Critical slowing down





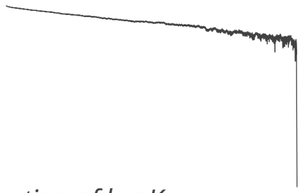
01. Arrhenius rejected
τ grows faster than exponential: ΔAIC = 11–26 for four τ definitions

02. $f_c \approx 1.8 – 1.95$
the critical force lies above every arresting run

03. γ ≈ 1.4 – 2.1
power law ≈ VFT; γ and f_c trade off against each other

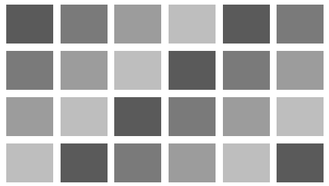
HAVOK: linear dynamics + rare kicks

1 Signal



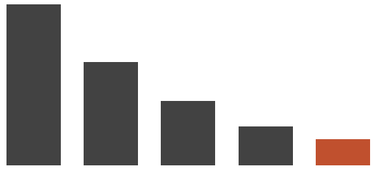
fluctuation of log K

2 Delay matrix



stack 20 t.u. of history

3 SVD



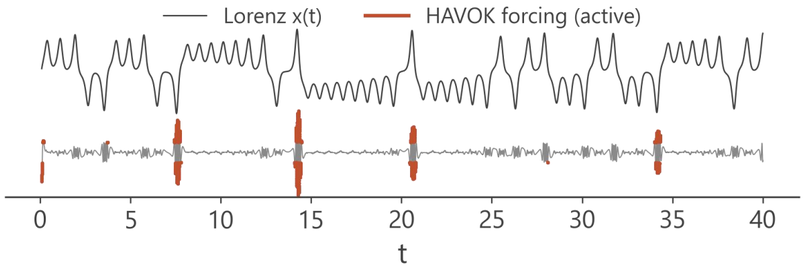
a few dominant patterns $v_1 \dots v_r$

4 Linear model

v_r quiet → linear

v_r bursts → something sudden

$dv/dt = A\,v + B\,v_r$

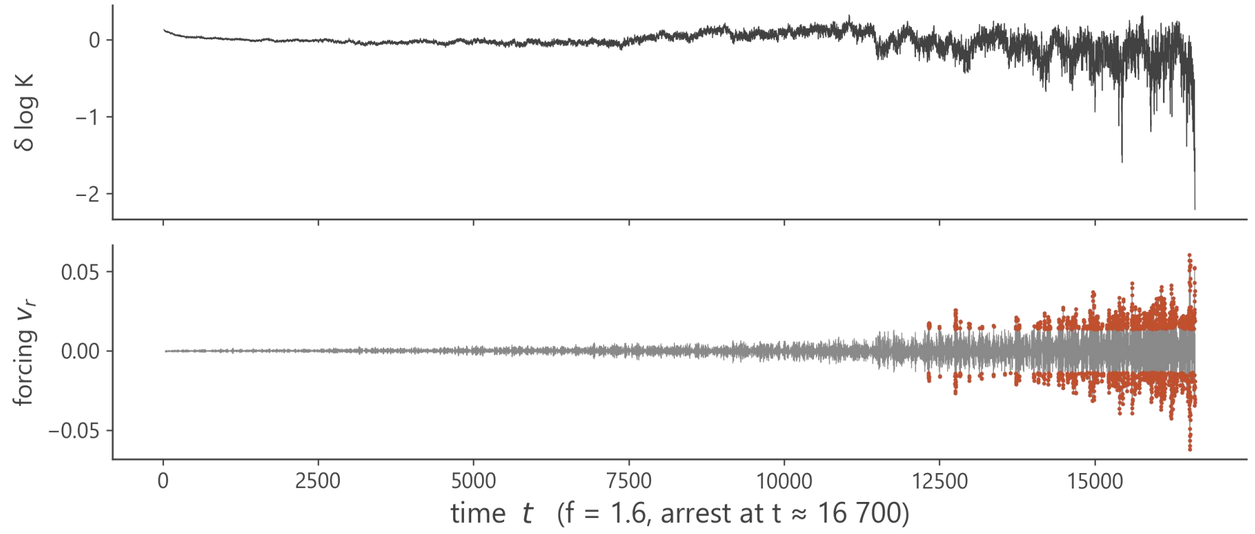


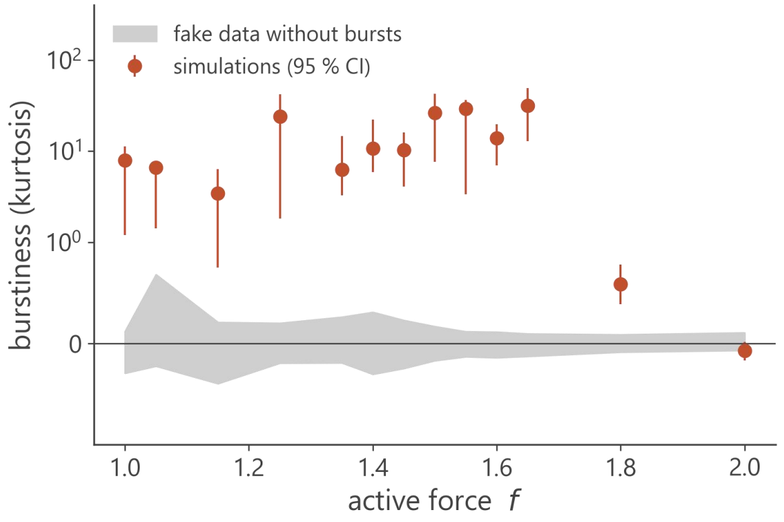
Check 1 · Lorenz system: the forcing fires at the lobe switches (2.7× chance).

Check 2 · PyDMD: the open-source implementation gives the same eigenvalue (0.189i vs 0.188i).

Our own implementation of Brunton et al., Nat. Commun. 8, 19 (2017).

Bursts announce the arrest





01.

Bursty only when arresting

kurtosis 3.5–32 for every arresting force; Gaussian at $f = 2.0$

02.

Real, not an artefact

above all 19 burst-free surrogates; same for 3 window sizes

03.

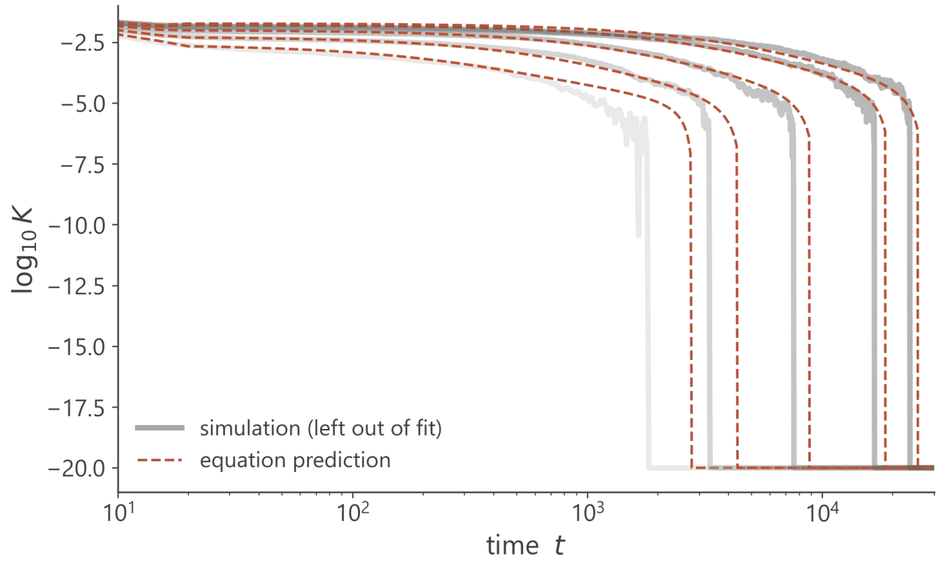
An early warning

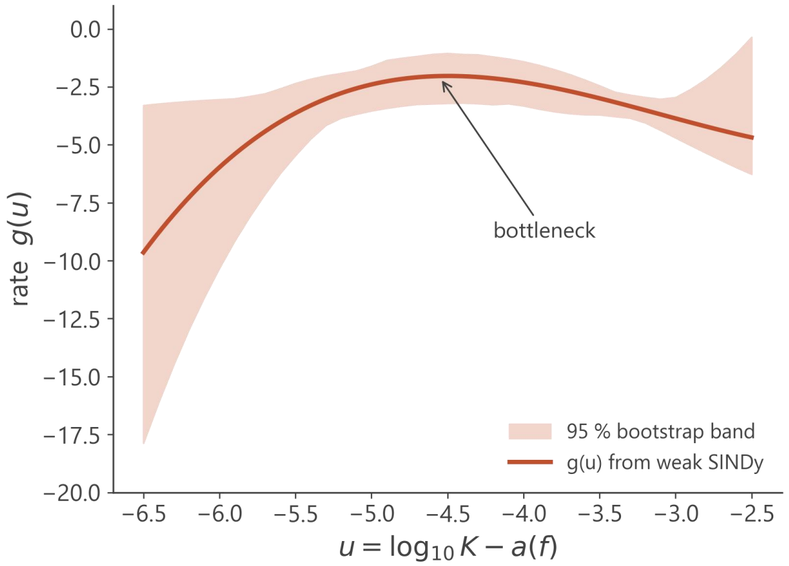
switches on in the last ~20 % and precedes sudden drops 2.9× more often than chance

A reduced equation for all forces

$$\tau(f) \cdot dy/dt = g(y - a(f)), \qquad y = \log_{10} K$$

$g(u) = 7.10u + 2.86u^2 + 0.307u^3$ *$\tau(f) = 1433\,(1.935 - f)^{-2.10}$* *then collapse at 0.40 decades / t.u.*





Weak-form SINDy · a generic polynomial library fails · leave-one-force-out arrest error 17 % · the differential form of the master curve, not a first-principles law

Q2 · Predicting runs never seen

(a) Unseen force

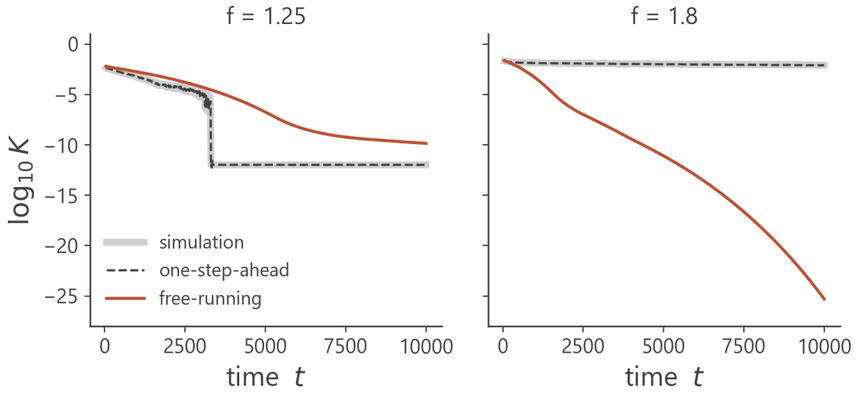
whole trajectory predicted from f alone

(b) Given the first part

10 / 25 / 50 % | *forecast the rest*

Five models, all rebuilt without the test run

- Neighbour baseline
- Parametric HDMD
- Deep Koopman AE
- Master-curve Koopman
- Discovered equation

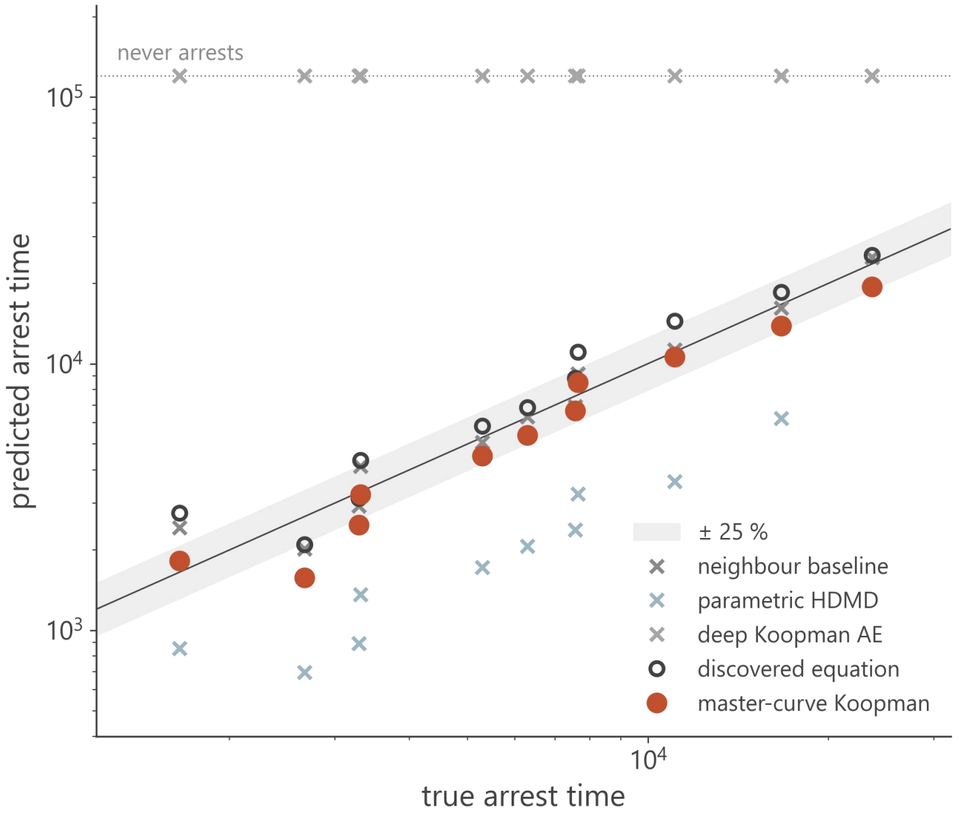


One-step-ahead ≠ forecasting.

The same model is 0.02 decades off when shown the true history every step, but 3.5–11.7 decades off when running on its own.

All results here are free-running and leave-one-force-out.

Q2 · Forecasting an unseen force



model	median error	missed	false
Neighbour baseline	8% [3%–24%]	0/11	1/2
Parametric HDMD	67% [59%–73%]	1/11	0/2
Deep Koopman AE	100% [100%–100%]	11/11	0/2
Master-curve Koopman	**14% [10%–18%]**	**0/11**	**0/2**
Discovered equation	17% [9%–31%]	0/11	0/2

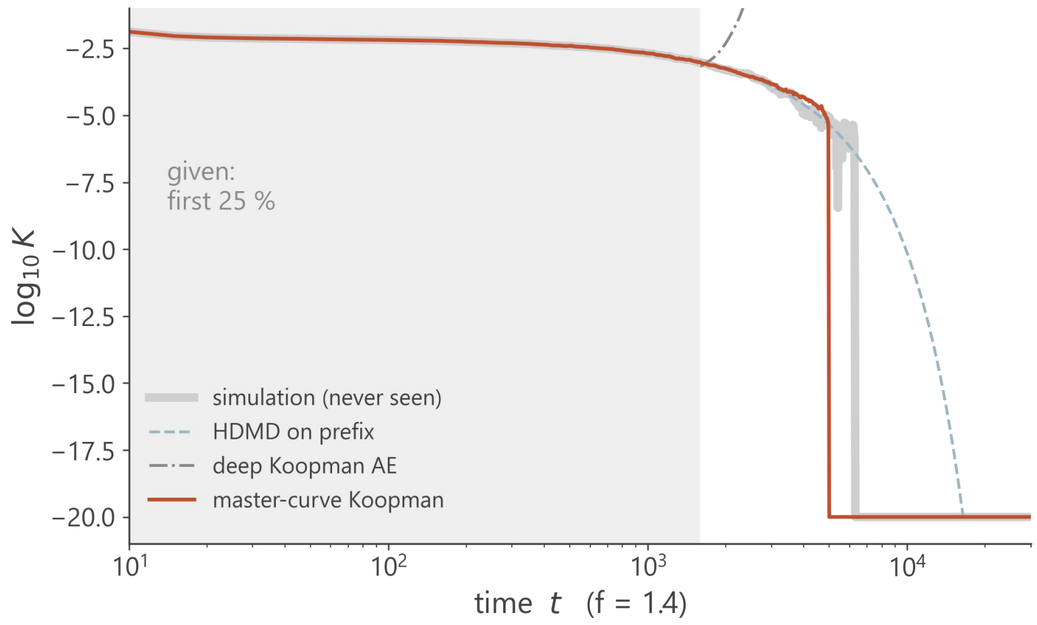
arrest-time error, 11 arresting forces; false arrests for f = 1.8, 2.0

Linear Koopman operators cannot create the abrupt arrest; the deep Koopman AE drifts.

Scaling-informed Koopman: **14%**, no missed or false arrests.

A time-stretched neighbour copy is at least as good inside the range (8%) but fails at the edges.

Q2 · Forecasting from the first part of a run



Master-curve Koopman, median arrest-time error

10 % given	*25 % given*	*50 % given*
20%	**15%**	**13%**

HDMD fitted on the prefix (≈ HAVOK without future kicks) and the deep Koopman AE miss most arrests (up to 6 and 11 of 11).

Discovered equation: 60% → 31%.

No missed arrests for the master-curve model at any prefix length.

How we tested every claim

claim	test that could have failed	result	
f = 1.65 arrests; f = 1.8 still decaying	block-bootstrap test of the slope	$p < 0.001$	●
one master curve for f = 1.0 – 1.8	predict arrest from slow decay alone; held-out forces	$R^2 = 0.97$, ratio ±15 %	●
τ diverges faster than exponential	AIC against Arrhenius, four τ definitions	ΔAIC = 11–26	●
$f_c \approx 1.8 – 1.95$, $\gamma \approx 1.4 – 2.1$	profile, bootstrap, fit windows	range only (degenerate)	●
intermittent bursts are real	19 burst-free surrogates; 3 window sizes	above all surrogates	●
reduced equation predicts unseen forces	leave-one-out; synthetic recovery; bootstrap	17 % error	●
$dK/dt = -AK + B f K^n$	fit n and the sign of B	unphysical ($B < 0$)	●
Koopman + master curve forecasts	free-running, leave-one-out, vs baseline	14%; ties baseline inside	●

● supported ● partly ● rejected

full list of 15 claims in the technical report: 10 supported · 4 partly · 1 rejected

Take-home messages

01. **Q1 · Scaling**

One master curve $\log_{10}K = a(f) + G(t/\tau(f))$; τ diverges near $f_c \approx 1.8–1.95$.

02. **Q2 · Prediction**

Plain Koopman models fail when run freely; a scaling-informed Koopman model predicts unseen runs (14%).

03. **Equation**

$\tau(f) \cdot dy/dt = g(y - a(f))$: compact, predictive, data-driven.

04. **HAVOK**

Intermittent bursts switch on before arrest: an early-warning signal.

Next steps

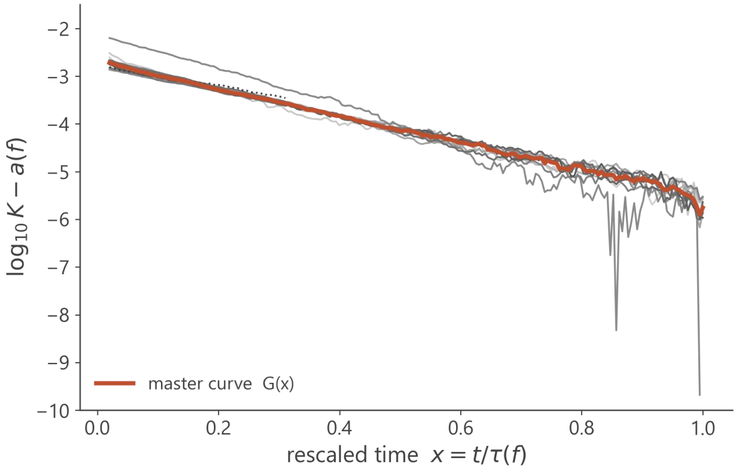
Several runs per force → error bars

Blind test: predict f = 1.7, 1.75, 1.85, 1.9 before simulating them

A second system size and density

Compare γ and G with active mode-coupling theory

Thank You



Decoding Dense Active Fluids Through Machine Learning

Questions are very welcome

Presented by :
Team B-9
Rithvik Arulprakash · Harshith KV · Vipin Sudhakar · Vijay Murugan

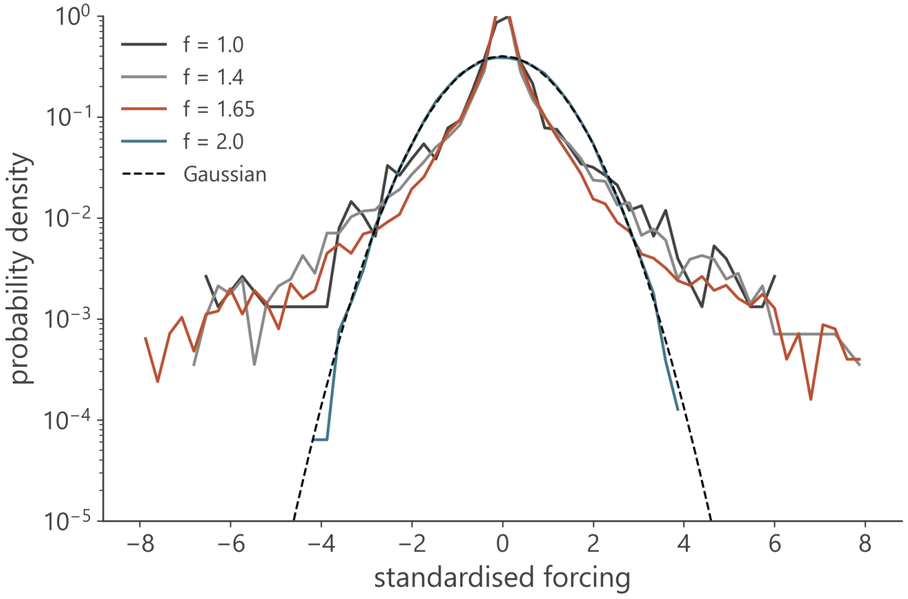
Code & data :
github.com/Rithvik188/Compflu

Backup · Critical fits

τ definition	best f_c	best γ	acceptable f_c (ΔAIC < 2)	acceptable γ	ΔAIC Arrhenius	ΔAIC VFT
arrest time	1.81	1.44	1.74 – 1.94	1.15 – 1.97	+11.1	+0.8
time to $K = 10^{-5}$	1.85	1.99	1.78 – 2.00	1.60 – 2.69	+12.0	+0.7
slow-decay time τ_K	1.89	1.89	1.86 – 1.93	1.66 – 2.18	+24.2	-2.3
master-curve τ	1.93	2.10	1.89 – 2.00	1.81 – 2.49	+20.4	-0.9
time to $K = 10^{-4}$ (biased)	2.02	3.08	1.92 – 2.17	2.58 – 3.85	+16.8	+1.2
time to $K = 10^{-3}$ (biased)	3.00	10.66	2.53 – 3.00	7.57 – 10.66	-6.7	+5.0
HDMD on linear K (biased)	3.00	6.40	2.60 – 3.00	4.83 – 6.40	-6.0	+6.6

$\tau = A\,(f_c - f)^{-\gamma}$ with f_c above every fitted force. Times to reach a fixed level mix the f-dependent offset a(f) with the time scale and are therefore biased.

Backup · HAVOK robustness



delay window	median kurtosis (arresting)	f = 1.8	f = 2.0	drops: hit / chance
10 t.u.	11.7	0.71	-0.00	3.3×
20 t.u.	10.7	0.58	-0.07	2.9×
40 t.u.	9.8	0.46	-0.03	3.1×

Every arresting force has fat-tailed forcing at every window size; f = 2.0 stays Gaussian.

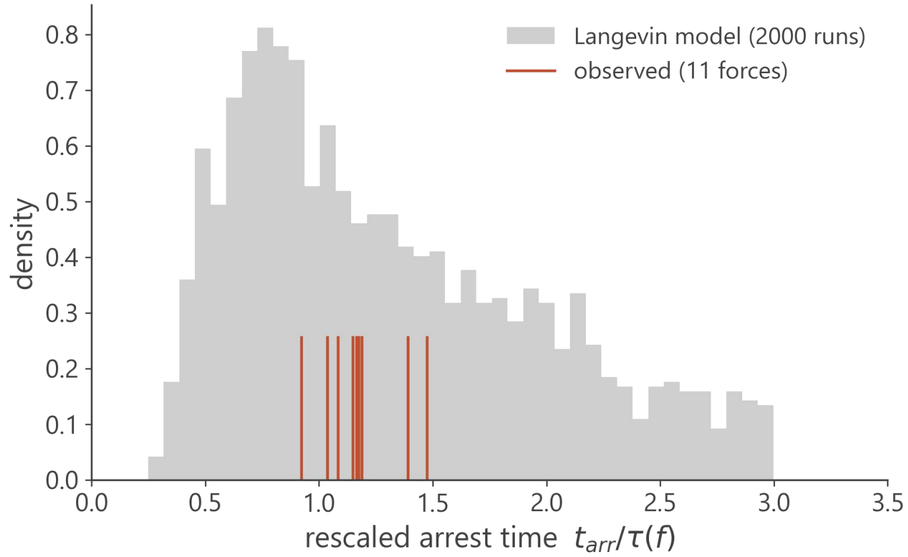
Surrogates: phase-randomised copies with the same power spectrum (Theiler et al., Physica D 1992).

Backup · Equation details

term	coefficient	95 % bootstrap CI	selected
1	0.000	[-12.02, 64.80]	18 %
u	7.102	[-4.83, 50.00]	99 %
u^2	2.859	[-0.62, 11.73]	98 %
u^3	0.307	[0.00, 0.89]	89 %

Polynomial degree (leave-one-out arrest error): quadratic 37%, cubic 17%, quartic 14%.

Synthetic test: the correct law is recovered up to twice the data's noise level.



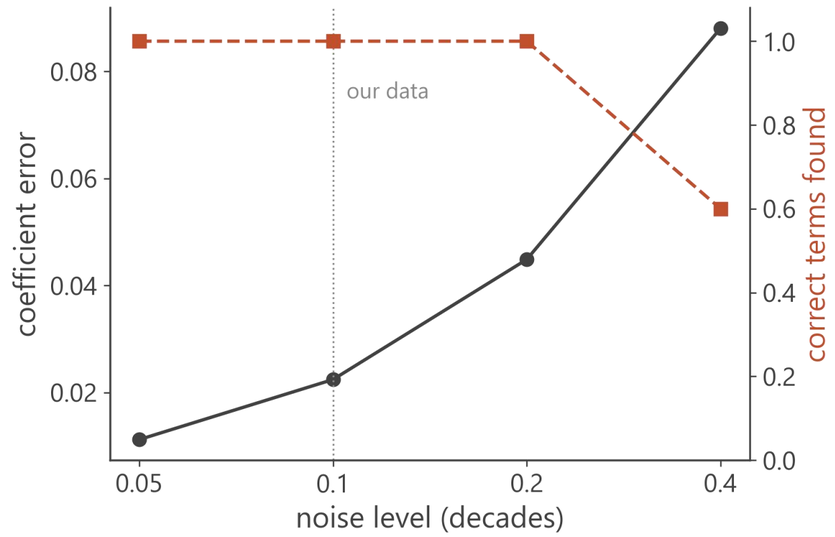
Langevin version: median rescaled arrest time 1.21 vs 1.17 observed; the spread is overestimated.

Backup · Per-force forecast errors

f	Neighbour	Param. HDMD	Deep Koopman	Master curve	Equation
1.0	+47 %	-48 %	no arrest	+10 %	+67 %
1.05	-24 %	-74 %	no arrest	-41 %	-22 %
1.15	-11 %	-73 %	no arrest	-24 %	-5 %
1.25	+25 %	-59 %	no arrest	-2 %	+31 %
1.35	-4 %	-67 %	no arrest	-14 %	+11 %
1.4	+0 %	-67 %	no arrest	-14 %	+9 %
1.45	-8 %	-69 %	no arrest	-12 %	+17 %
1.5	+20 %	-57 %	no arrest	+11 %	+45 %
1.55	+2 %	-67 %	no arrest	-4 %	+31 %
1.6	-3 %	-63 %	no arrest	-17 %	+11 %
1.65	+6 %	no arrest	no arrest	-18 %	+8 %
1.8	false arrest	correct	correct	correct	correct
2.0	correct	correct	correct	correct	correct

relative error of the predicted arrest time; f = 1.0 and f ≥ 1.65 require extrapolation in f

Backup · Method checks



Weak SINDy recovers the right terms on synthetic data with the simulations' correlated noise up to 0.2 decades (ours ≈ 0.1).

τ_K fit window	γ	f_c	R^2
t = 20 … 0.70 t_{arr}	1.94	1.893	0.988
t = 20 … 0.85 t_{arr}	1.89	1.889	0.984
t = 20 … 0.95 t_{arr}	1.89	1.889	0.978
t = 100 … 0.70 t_{arr}	1.92	1.891	0.987
t = 100 … 0.85 t_{arr}	1.87	1.887	0.982
t = 100 … 0.95 t_{arr}	1.87	1.887	0.977

γ and f_c barely move when the slow-stage window is changed.